

Faculty of Computer Technology and Cybersecurity
Department of Computer Engineering
ARCHITECTURE AND ORGANIZATION OF COMPUTER SYSTEMS
WORKSHEET - 1

Instructor: Dr. Mohsin Farhad, PhD, Professor

Points: 0.90

Date: 05.09.2026

Full Name: Daniil Kalts

Group: IT2-2404SE

Q1: Describe Computer Architecture

Computer architecture is the main design of a computer system. It describes what the computer can do and how software can use it. It includes the instruction set, data types, memory structure, input/output operations, and addressing methods.

Q2: Explain the structure and role of the processor and its components

The processor, also called the CPU, is the main part of a computer. It executes instructions and processes data. Its main components are the Control Unit, the Arithmetic Logic Unit, and registers.

The Control Unit controls the work of the CPU. It fetches and decodes instructions and sends control signals to other components. The Arithmetic Logic Unit performs arithmetic and logical operations, such as addition, subtraction, AND, OR, and comparison. Registers are very small and fast storage locations inside the CPU. They store data, instructions, addresses, and temporary results while a program is running.

Q3: What are some technological developments in Computer Architecture?

Computer architecture has changed a lot over time. Early computers used vacuum tubes. Later, engineers developed transistors, integrated circuits, and microprocessors. These technologies made computers smaller, faster, cheaper, and more reliable.

Modern computers use multi-core processors, cache memory, pipelining, and parallel processing. They can also use special processors, such as graphics processing units (GPUs) and AI accelerators. These technologies help computers process large amounts of data faster.

Q4: What is the difference between Computer Architecture and Computer Organization?

Computer architecture explains what a computer can do from the programmer's view. It includes instructions, data types, addressing methods, and input/output operations.

Computer organization explains how the computer is built to perform these functions. It includes hardware details, such as registers, buses, cache memory, pipelines, and connections between components.

For example, architecture defines an instruction that the CPU can use. Organization explains how the hardware performs this instruction.

Q5: Explain the different types of data manipulation

Data manipulation means working with data to change, organize, or use it. The main types are:

1. **Arithmetic operations** - addition, subtraction, multiplication, and division.
2. **Logical operations** - AND, OR, NOT, and XOR.
3. **Comparison operations** - checking if values are equal, greater, or smaller.
4. **Data transfer** - moving data between registers, memory, and input/output devices.
5. **Data conversion** - changing data from one format or type to another.
6. **Sorting and searching** - putting data in order and finding needed information.